

modbus protocol			
From:	To:		
Vesion	V1.0	update:2023.06.21	

1. Transmission Format

Baud Rate	9600bps
Start bit	1
Byte width	8
Parity	N
Stop bits	1
Slave address	H10

2. Packet Format

Address	Function	Data	CRC checksum
16bits	16bits 03:Function of reading multi registers 16:Function of presenting multi registers	N*16bits	16bits

3. Data types

Data Types	Description
TEMP1	have symble byte, 0.1℃resolution, formula: T*10, temperature range :-30~97℃ (When the temperature show 25 ℃, the protocol data transmission is 250;When the temperature show -25 ℃, the protocol data transmission is -250 ; when bit15 is 1 represents the negative number, when the bit15 is 0, it represents Integer;) , When the value is 32767,it represents the sensor failure
DIGI1	No symble byte, unit :1, When show 123,the data transmission is 123;
DIGI2	No symble byte, unit :10, When show 1230,the data transmission is 123;
DIGI3	No symble byte, unit :100, When show 12300,the data transmission is 123;
DIGI4	No symble byte, unit :5, When show 10,the data transmission is 2;
DIGI5	No symble byte, unit :0.1, When show 12.3,the data transmission is 123;
DIGI7	No symble byte, unit :0.01, When show 1.23,the data transmission is 123;
DIGI6	No symble byte, unit :0.001, When show 0.123,the data transmission is 123;

4. Mailing address

Address	Function	Number	Content	Byte length	mode	Description	Remark
5001	03/06		ON/OFF	double byte	read/write	0-OFF/1-ON	DIGI1
5002	03/06		Cooling Target Temp.	double byte	read/write		TEMP
5003	03/06		Heating Target Temp.	double byte	read/write		TEMP
5005	03/06		Mode	double byte	read/write	0-Cooling/1-Heating	DIGI1
5028	03/06		Working mode of circulating water pump	double byte	read/write	0-normal/1-special/2-intermittent	DIGI1
5029	03/06		Circulating water pump running time interval	double byte	read/write	1~120min	DIGI1

5030	03/06		Circulating water pump running duration	double byte	read/write	1~30min	DIGI1
5031	03/06		Circulating water pump advance compressor	double byte	read/write	0~30min	DIGI1
5033	03/06		Temp. Diff. for Power-on in Heating	double byte	read/write	0~10℃	TEMP
5034	03/06		Temp. Diff. for Power-on in Cooling	double byte	read/write	0~10℃	TEMP
5055	03/06		Whether to clear 3 or more lock faults	double byte	read/write	0-No/1-Yes	DIGI1
2001							
2002							
2003							
2004							
2005							
2006							
2007							
2008							
2009							
2010							
2011	03		Unit state	double byte	read	0-Off/1-On	DIGI1
2012	03		Unit mode	double byte	read	0-Cooling/1-Heating/250-Defrost	DIGI1
2013	03		Current set temperature value	double byte	read		TEMP
2016	03		Inlet water temp.	double byte	read		TEMP
2017	03		Outlet water temp.	double byte	read		TEMP
2019	03		Ambient temp.	double byte	read		TEMP
2040	03		Sysrem 1 Operation Frequency of Compressor	double byte	read	0~200Hz	DIGI1
2041	03		Sysrem 2 Operation Frequency of Compressor	double byte	read	0~200Hz	DIGI1
2044	03		Sysrem 1 Speed of Fan Motor	double byte	read	0~1300r	DIGI1
2069	03		bit0:System 1 High pressure switch(0-closed/1-open) bit1:System 2 High pressure switch(0-closed/1-open) bit4:System 1 Low pressure switch(0-closed/1-open) bit5:System 2 Low pressure switch(0-closed/1-open) bit8:Flow switch	double byte	read		DIGI1
2070	03		Sysrem 1 Fan Motor	double byte	read	0-Close/1-High speed/2-Low speed	DIGI1
2072	03		bit0:Compressor 1 output (0-OFF/1-ON) bit1:Compressor 2 output (0-OFF/1-ON) bit4:Fan motor 1 output (0-OFF/1-ON) bit8:Circulating water pump output (0-OFF/1-ON) bit9:Pan heater (0-OFF/1-ON) bit11:Electric heater output (0-OFF/1-ON) bit12:System 1 Crankcase Heater output (0-OFF/1-ON) bit13:System 2 Crankcase Heater output (0-OFF/1-ON)	double byte	read		DIGI1
2073	03		bit2:4 way valve 1 output (0-OFF/1-ON) bit3:4 way valve 2 output (0-OFF/1-ON) bit7:alarm output (0-OFF/1-ON)	double byte	read		DIGI1
2074	03		bit0:Inlet water temp. failure (0-no/1-yes) bit1:Outlet water temp. failure (0-no/1-yes) bit3:Ambient temp. failure (0-no/1-yes) bit4:System 1 Suction temp. failure (0-no/1-yes) bit5:System 2 Suction temp. failure (0-no/1-yes) bit8:System 1 Exhaust temp. failure (0-no/1-yes) bit9:System 2 Exhaust temp. failure (0-no/1-yes)	double byte	read		DIGI1

2075	03		bit4:System 1 Anti-freeze temp. failure (0-no/1-yes) bit5:System 2 Anti-freeze temp. failure (0-no/1-yes) bit8:System 1 coil 1 temp. failure (0-no/1-yes) bit9:System 1 coil 2 temp. failure (0-no/1-yes) bit10:System 2 coil 1 temp. failure (0-no/1-yes) bit11:System 2 coil 2 temp. failure (0-no/1-yes)	double byte	read		DIG11
2076	03		bit0:System 1 High pressure protection (0-no/1-yes) bit1:System 2 High pressure protection (0-no/1-yes) bit4:System 1 Low pressure protection (0-no/1-yes) bit5:System 2 Low pressure protection (0-no/1-yes) bit8:System 1 Anti-freeze protection (0-no/1-yes) bit9:System 2 Anti-freeze protection (0-no/1-yes) bit12:System 1 Exhaust over heat protection (0-no/1-yes) bit13:System 2 Exhaust over heat protection (0-no/1-yes)	double byte	read		DIG11
2077	03		bit0:Syetem 1 High pressure sensor failure (0-no/1-yes) bit1:Syetem 2 High pressure sensor failure (0-no/1-yes) bit4:Syetem 1 Low pressure sensor failure (0-no/1-yes) bit5:Syetem 2 Low pressure sensor failure (0-no/1-yes) bit12:System 1 four-way valve switching abnormality bit13:System 2 four-way valve switching abnormality	double byte	read		DIG11
2078	03		bit0:Winter primary anti-freeze protection (0-no/1-yes) bit1:Winter secondary class anti-freeze protection (0-no/1-yes) bit2:Outlet water temp. over heat protection(0-no/1-yes) bit3:Outlet water temp. over low protection(0-no/1-yes) bit4:Low ambient temperature protection(0-no/1-yes) bit5:Electric heating overload protection (0-no/1-yes) bit6:Water flow protection (0-no/1-yes) bit7:The temp. difference of inlet water and outlet water over large protection (0-no/1-yes) bit10:Abnormal power failure protection (0-no/1-yes) bit11:Communication failure(main board with fan motor module board) (0-no/1-yes) bit13:Low ambient temperature does not allow cooling protection (0-no/1-yes)	double byte	read		DIG11
2079	03		bit0:System 1 Exhaust over heat protection 3 times (0-no/1-yes) bit1:System 2 Exhaust over heat protection 3 times (0-no/1-yes) bit4:SyStem 1 high pressure protection 3 times (0-no/1-yes) bit5:SyStem 2 high pressure protection 3 times (0-no/1-yes) bit8:SyStem 1 low pressure protection 3 times (0-no/1-yes) bit9:SyStem 2 low pressure protection 3 times (0-no/1-yes) bit12:SyStem 1 Anti-freeze protection 3 times (0-no/1-yes) bit13:SyStem 2 Anti-freeze protection 3 times (0-no/1-yes)	double byte	read		DIG11

2080	03		bit0:Outlet water overtemp protection 3 times (0-no/1-yes) bit1:Outlet water too low protection 3 times (0-no/1-yes) bit2:Electric heater over heat protection 3 times (0-no/1-yes) bit3:Water flow protection 3 times (0-no/1-yes) bit4:The temp difference of inlet water and outlet water temp over large 3 times (0-no/1-yes) bit6:Fan motor 1 overload protection 3 times (0-no/1-yes) bit10:Fan motor 1 overload protection (0-no/1-yes)	double byte	read		DIGI1
2081	03		bit0:F151 System 1 inverter board communication failure bit1:F152 System 1 compressor start failure bit2:F153 System 1 compressor starts IPM protection bit3:F154 System 1 compressor running IPM protection bit4:F156 System 1 compressor overcurrent protection bit5:F155 System 1 IPM over temp. protection bit6:F157 System 1 DC bus overvoltage protection bit7:F158 System 1 DC bus undervoltage protection bit8:F251 System 2 inverter board communication failure bit9:F252 System 2 compressor start failure bit10:F253 System 2 compressor starts IPM protection bit11:F254 System 2 compressor running IPM protection bit12:F256 System 2 compressor overcurrent protection bit13:F255 System 2 IPM over temp. protection bit14:F257 System 2 DC bus overvoltage protection bit15:F258 System 2 DC bus undervoltage protection	double byte	read		DIGI1
2083	03		bit0:F101 Fan 1 output phase loss protection bit1:F102 Fan 1 output no speed protection bit2:F103 Fan 1 starts IPM protection bit3:F104 Fan 1 running IPM protection bit4:F105 Fan 1 over current protection bit5:F106 Fan 1 over heat protection bit6:F107 Fan 1 DC bus overvoltage protection bit7:F108 Fan 1 DC bus undervoltage protection bit8:F201 Fan 2 output phase loss protection bit9:F202 Fan 2 output no speed protection bit10:F203 Fan 2 starts IPM protection bit11:F204 Fan 2 running IPM protection bit12:F205 Fan 2 over current protection bit13:F206 Fan 2 over heat protection bit14:F207 Fan 2 DC bus overvoltage protection bit15:F208 Fan 2 DC bus undervoltage protection	double byte	read		DIGI1
2085	03		bit0:F112 Fan 1 current sampling failure bit1:F212 Fan 2 current sampling failure bit4:F109 Fan 1 overspeed protection bit5:F209 Fan 2 overspeed protection bit8:F110 Fan 1 stall protection bit9:F210 Fan 2 stall protection bit12:F088 Compressor Type Error	double byte	read		DIGI1

2086	03		bit0:E064 The temp. difference of inlet water and outlet water abnormality protection bit1:E064-3 The temp. difference of inlet water and outlet water abnormality protection 3 times bit4:E121-3 System 1 four-way valve switching abnormality 3 times bit5:E221-3 System 2 four-way valve switching abnormality 3 times	double byte	read		
2087	03			double byte	read		
2088	03			double byte	read		
2089	03			double byte	read		
2090	03			double byte	read		